

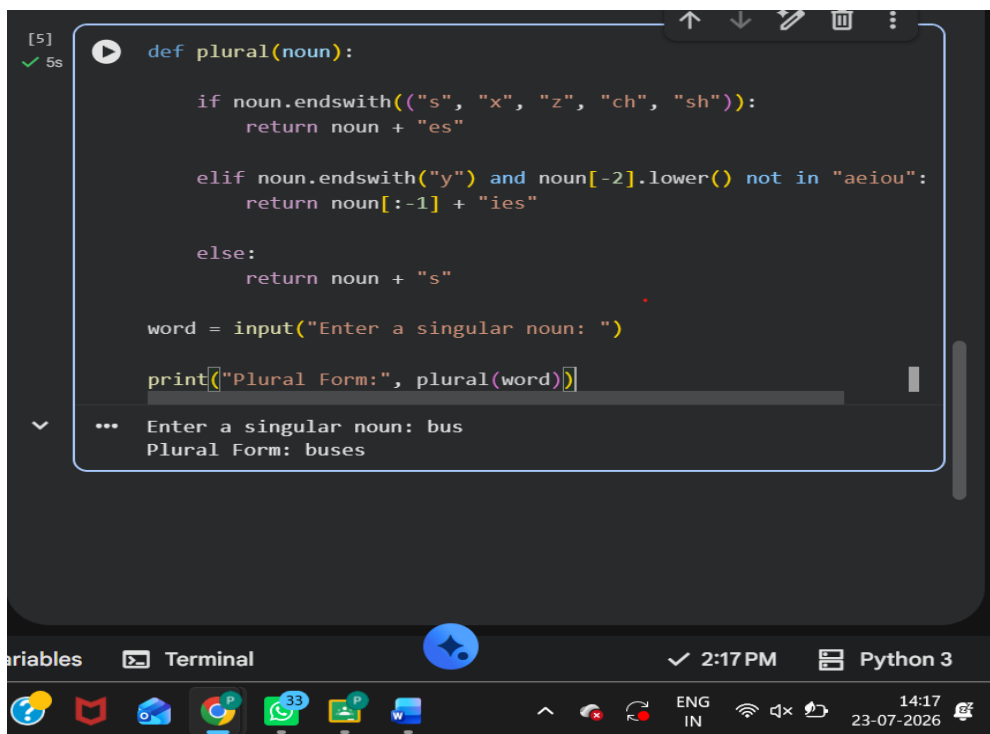
Experiment 4 :

Implement a finite-state machine for morphological parsing. In this example, generate plural forms of English nouns using Python.

Code:

```
def plural(noun):  
    if noun.endswith(("s", "x", "z", "ch", "sh")):  
        return noun + "es"  
  
    elif noun.endswith("y") and noun[-2].lower() not in "aeiou":  
        return noun[:-1] + "ies"  
  
    else:  
        return noun + "s"  
  
word = input("Enter a singular noun: ")  
print("Plural Form:", plural(word))
```

Output



```
[5]  
✓ 5s  
def plural(noun):  
    if noun.endswith(("s", "x", "z", "ch", "sh")):  
        return noun + "es"  
  
    elif noun.endswith("y") and noun[-2].lower() not in "aeiou":  
        return noun[:-1] + "ies"  
  
    else:  
        return noun + "s"  
  
    word = input("Enter a singular noun: ")  
  
    print("Plural Form:", plural(word))  
... Enter a singular noun: bus  
Plural Form: buses
```

Variables Terminal Python 3 2:17 PM

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